

RHYTHM GILHOTRA

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OBJECTIVE

Data Engineer with 4+ years of experience designing large-scale, real-time data pipelines, distributed ETL systems, and cloud-native analytical platforms on GCP. Adept at translating business requirements into scalable data architectures, enabling BI, analytics, and decision intelligence across global financial systems. Skilled in stakeholder collaboration, technical workshops, data modeling, and modern DevOps practices to deliver future-proof data solutions.

TECHNICAL SKILLS

Programming	Python, Java, Kotlin, SQL, Bash
Data Engineering	Streaming Pipelines, ETL/ELT, Data Modeling, Data Warehousing
Cloud Platforms	Google Cloud (GCP), BigQuery, Dataflow, Pub/Sub, Kubernetes, Docker
Tools	Airflow, Spark, Kafka, Hive, Looker, Grafana, Prometheus, Git
Analytics & BI	KPI Frameworks, Dashboarding (Looker), A/B Testing, Data Analysis
DevOps	GitHub Actions, Cloud Build, CI/CD, Terraform, Observability, Logging
Consulting Skills	Requirement Gathering, Technical Workshops, Documentation, Stakeholder Management

EXPERIENCE

Data Engineer II American Express	May 2023 – Present <i>Gurugram, India</i>
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- Designed and deployed real-time streaming pipelines on GCP using Pub/Sub and Dataflow, powering credit-risk decisioning systems and improving model signal freshness by 70%.
- Led requirement-gathering and solution design workshops with product, risk, and analytics teams, translating business pain points into actionable data engineering designs.
- Built analytics-ready datasets in BigQuery to support BI, KPI tracking, and operational dashboards used across global fraud and risk operations.
- Implemented CI/CD workflows with GitHub Actions and Cloud Build for automated deployment of pipelines and transformations.
- Improved platform observability through structured logging, Prometheus metrics, and Grafana dashboards, reducing incident detection time by 40%.
- Collaborated with cross-functional and distributed teams (onshore/offshore) to deliver scalable solutions aligned with strategic risk-intelligence goals.

Data Engineer I American Express	Aug 2021 – May 2023 <i>Gurugram, India</i>
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- Built distributed ETL pipelines using Airflow and BigQuery, optimizing compute workloads and reducing processing time from 6 hours to 90 minutes.
- Partnered with business and analytics stakeholders to understand reporting needs and translate them into scalable data models and transformation pipelines.
- Developed aggregation layers for Looker dashboards via BigQuery BI Engine, reducing fraud investigation turnaround time by 40%.
- Designed SLA-based monitoring, anomaly detection, and automated recovery pipelines ensuring 99.9% availability for regulatory and operational datasets.
- Created high-quality technical documentation, lineage diagrams, and architecture specifications to support cross-team adoption and platform onboarding.

Data Engineering Intern

American Express

Jan 2021 – Jun 2021

Remote

- Built real-time risk dashboards and reporting pipelines using Python and SQL, improving data freshness by 60%.
- Developed automated data validation and profiling scripts to improve trust and reliability in operational reporting.
- Conducted exploratory data analysis to support fraud detection initiatives and insight generation.

PROJECTS

Distributed Network Intrusion Detection System (NIDS)

Thapar Institute of Engineering and Technology

- Designed a distributed data processing system analyzing 10K+ packets/sec using Python and TensorFlow for real-time anomaly detection.
- Created ingestion and alerting modules with modular APIs, reducing false positives by 20%.
- Deployed the system via Docker with full observability support using Grafana.

Scalable Transaction Data Pipeline

American Express

- Engineered a large-scale data pipeline using Hadoop, Spark, and Kafka to process millions of transaction records daily for fraud analytics.
- Built fault-tolerant ingestion and transformation layers with SLA monitoring and retries, improving data freshness by 40%.

EDUCATION

Bachelor of Engineering in Computer Engineering, *Thapar University, Patiala, India*

2017 – 2021

Relevant Coursework: Distributed Systems, Data Engineering, Database Systems, Statistics, Algorithms

LEADERSHIP

Tech Lead & Mentor, American Express

2023 – Present

Mentored 4+ engineers on scalable data architecture, observability, and best practices in data engineering and distributed systems.

Student Leader, Thapar Institute of Engineering and Technology

2019 – 2021

Led technical workshops on data systems, distributed architecture, and engineering foundations.

AWARDS

Star Award, American Express

Q1 2025

Recognized for leading deployment of real-time fraud intelligence platforms and improving global risk data availability.

Analyst of the Quarter, American Express

Q1 2024, Q3 2022

Awarded for delivering high-impact data engineering solutions and optimizing mission-critical analytical pipelines.